High-Yield Revision Sheet: Atmospheric Circulation & Weather Systems

Subject: Climatology & Physical Geography | NCERT Class 7 Ch. 4 & Class 11 Ch. 9 | UPSC & State PSC Core

⚡ 30-Second Snapshot

- Atmospheric pressure decreases with altitude from an average sea-level standard of **1,013.2 mb**, driving winds from high to low pressure zones.
- Wind velocity and direction are governed by three primary forces: **Pressure Gradient Force**, **Frictional Drag**, and the **Coriolis Force** (which deflects winds right in the North and left in the South, and is zero at the equator).
- Global pressure is organized into permanent belts: **Equatorial Low**, **Sub-Tropical Highs**, **Sub-Polar Lows**, and **Polar Highs**, linked by the **Hadley, Ferrel, and Polar circulation cells**.
- **Air masses** acquire uniform temperature and moisture over homogeneous source regions, creating active boundaries (**fronts**) when they collide.
- **Extra-tropical cyclones** form along polar fronts in mid-latitudes with clear frontal systems, whereas **tropical cyclones** draw energy exclusively from warm ocean waters (>27°C) and feature a calm central **eye** and destructive **eye wall**.

🏛 Comparative Matrix: Global Wind Systems & Cyclones

Atmospheric System	Spatial Extent & Location	Core Driving Mechanism	Weather Characteristics & Impacts
Trade Winds	Tropics (0° to 30° latitude)	Surface return flow from Subtropical Highs to Equatorial Low.	Steady easterly winds that drive tropical ocean currents.
Prevailing Westerlies	Mid-Latitudes (30° to 60°)	Surface flow from Subtropical Highs to Sub-Polar Lows.	Persistent westerly winds dominating mid-latitude weather tracks.
Extra-Tropical Cyclones	Mid-to-High Latitudes	Polar front wave development along contrasting air masses.	Large areal coverage, lower wind speeds, clear cold/warm fronts, west-to-east movement.
Tropical Cyclones	Tropical Oceans (5° to 20° latitude)	Latent heat release from warm sea surface temperatures (>27°C).	Small diameter, extreme wind velocities (>250 km/h), torrential rain, storm surges, eye wall destruction.

📐 Wind Forces & Cyclone Structural Mechanics

- **Forces Affecting Wind:**
  - **Pressure Gradient Force (PGF):** Drives air from high to low pressure; strong where isobars are tightly packed.
  - **Coriolis Force:** Apparent deflection due to Earth's rotation; deflects wind right in the Northern Hemisphere, left in the Southern Hemisphere, and is zero at the equator.
  - **Geostrophic Wind:** Upper-air wind flowing parallel to straight isobars where PGF balances the Coriolis force.
- **Tropical Cyclone Anatomy & Dissipation:**
  - **The Eye:** A calm, 150–250 km wide central core characterized by subsiding air, light winds, and clear skies.
  - **The Eye Wall:** A dense ring surrounding the eye featuring towering cumulonimbus clouds, extreme updrafts, torrential rainfall, and peak destructive winds.
  - **Landfall & Decay:** When a tropical cyclone crosses a coastline, its moisture and latent heat supply are cut off, causing the storm to rapidly weaken and dissipate.

⚠ Common UPSC Exam Traps

- **Trap 1 (Coriolis Force at the Equator):** The Coriolis force is not uniform across the globe; it is strictly **zero at the equator** and reaches its maximum intensity at the poles. This is why tropical cyclones cannot form within 5 degrees of the equator.

- **Trap 2 (Tropical vs. Extra-Tropical Cyclones):** Extra-tropical cyclones are associated with distinct frontal systems (warm and cold fronts) and can form over land or sea. Tropical cyclones have **no fronts**, form exclusively over warm tropical oceans, and derive their energy entirely from latent heat of condensation.
- **Trap 3 (Cyclone Rotation Hemispheric Rule):** In the Northern Hemisphere, surface cyclones rotate **counterclockwise** (anticlockwise), whereas in the Southern Hemisphere, they rotate **clockwise**.
- **Trap 4 (Barchan vs. Dune Horns — Review):** As noted previously, barchan horns point downwind, while atmospheric anticyclones diverge outward at the surface.

🎯 High-Yield Facts Box

Meteorological Parameter	Standard Benchmark Value	Climatological Significance
Average Sea-Level Pressure	1,013.2 mb	Standard barometric baseline measured in millibars.
SST Threshold for Cyclones	Greater than 27°C	Minimum ocean water temperature required to sustain tropical cyclone formation.
Tropical Cyclone Eye Width	150 to 250 kilometers	Calm central atmospheric core surrounded by the destructive eye wall.
Coriolis Equator Rule	Zero deflection	Prevents rotational vortex formation very close to the equator.
Wind Deflection Direction	Right (North) / Left (South)	Hemispheric rule governing the deflection of moving air and water.

📝 Prelims Practice Challenge

Q1. Consider the following statements regarding the forces that affect wind speed and direction:

1. The Coriolis force deflects winds to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.
2. The magnitude of the Coriolis force is directly proportional to the angle of latitude and is maximum at the equator.
3. Geostrophic winds blow parallel to isobars in the upper atmosphere when the pressure gradient force is balanced by the Coriolis force.

Which of the statements given above are correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2, and 3

> **Answer: (C) 1 and 3 only**
> **Explanation:** Statement 1 is correct because the Coriolis force deflects winds to the right in the North and to the left in the South. Statement 2 is incorrect because the Coriolis force is **zero at the equator** and maximum at the poles, not maximum at the equator. Statement 3 is correct because upper-air geostrophic winds flow parallel to straight isobars under the balance of PGF and Coriolis force.

Q2. With reference to tropical cyclones, consider the following statements:

1. They require sea surface temperatures in excess of 27°C to provide continuous moisture and latent heat energy.
2. They feature a well-defined frontal system separating contrasting warm and cold air masses.
3. The central "eye" of a mature tropical cyclone is characterized by calm conditions, subsiding air, and clear skies.

Which of the statements given above is/are correct?

- (A) 1 and 3 only
- (B) 2 only
- (C) 1 and 2 only
- (D) 1, 2, and 3

> **Answer: (A) 1 and 3 only**
> **Explanation:** Statement 1 is correct because warm ocean waters above 27°C supply the necessary latent heat of condensation. Statement 2 is incorrect because tropical cyclones are **warm-core systems with no fronts**; frontal systems are characteristic of mid-latitude extra-tropical cyclones. Statement 3 is correct because the central eye is a calm region of descending air and clear weather.

✏️ Mains Practice Question (10 Marks / 150 Words)

Question: *"Analyze the genesis conditions necessary for the formation of tropical cyclones. Why are coastal regions bordering the Bay of Bengal particularly prone to severe tropical storm devastation?"*

💭 Structured Answer Framework:

- **Introduction (25–30 words):**
  Define tropical cyclones as violent, warm-core low-pressure storms originating over tropical oceans, driven by latent heat release from moisture condensation.
- **Body Arguments (80–90 words):**
  - *Genesis Conditions:* Formation requires sea surface temperatures exceeding 27°C, sufficient Coriolis force (away from the equator), low vertical wind shear, and pre-existing low-pressure vorticity.
  - *Bay of Bengal Vulnerability:* The Bay of Bengal receives high solar insolation, keeping sea surface temperatures consistently above the 27°C threshold. Its semi-enclosed basin traps heat and funnels storm surges toward densely populated low-lying coasts.
  - *Topographical & Societal Factors:* Shallow bathymetry, funnel-shaped coastal topography, high tidal ranges, and dense coastal populations exacerbate storm surges, resulting in massive loss of life and property (as seen in the 1999 Odisha Super Cyclone).
- **Conclusion / Way Forward (25–30 words):**
  Mitigating tropical cyclone devastation requires strengthening early warning radar networks, constructing cyclone shelters, and implementing coastal shelterbelt plantations.

✍ Notes & Updates